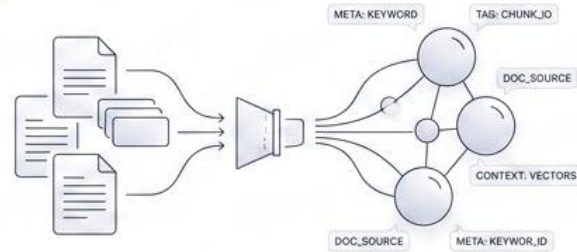


Chunk Enrichment & Vector Databases

Turning Raw Chunks into
Intelligent Data



What is Chunk Enrichment?

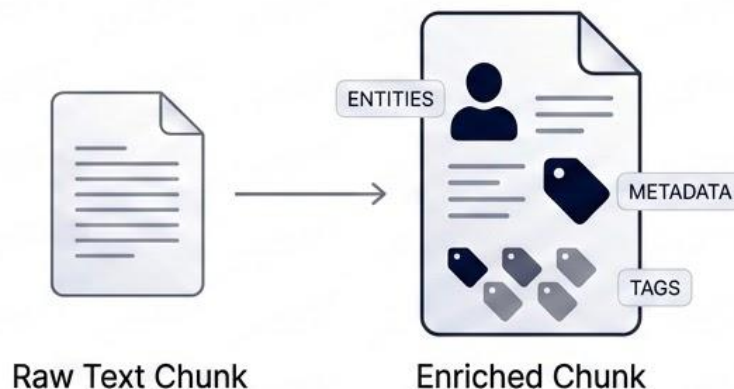
Enhancing raw text chunks with additional information to improve retrieval quality

Why it matters

- Improved semantic understanding.
- Reduces retrieval hallucination.
- Adds critical context.

Key Benefits

- 🔍 Accurate Retrieval
- 🔗 Context Retention
- 🏗️ Structured Organization
- 🏛️ Domain Specificity



PII Detection & Redaction

Personally Identifiable Information (PII)

- Name, Email, Phone, Aadhaar, PAN, etc.

Why redact: Privacy, Compliance

- GDPR (Global Data Protection Regulation)
- DPDP (Digital Personal Data Protection Act)

Techniques

- Regex (Regular Expressions)
- Microsoft Presidio
- LLMs (Large Language Models)

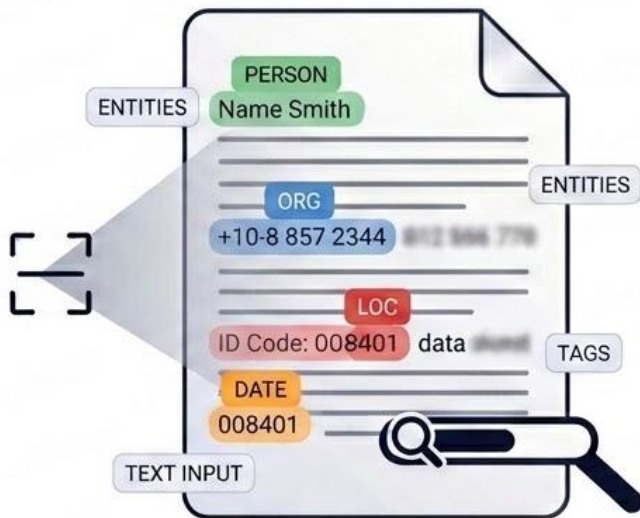


Named Entity Recognition (NER)

 **Definition:** Extracting key entities like Person, Organization, Location, Date, etc.

 **Benefits:**

- Better metadata
- Improved filtering & search



Key-Phrase Extraction & Summarization



Techniques

- TF-IDF; RAKE;
- KeyBERT



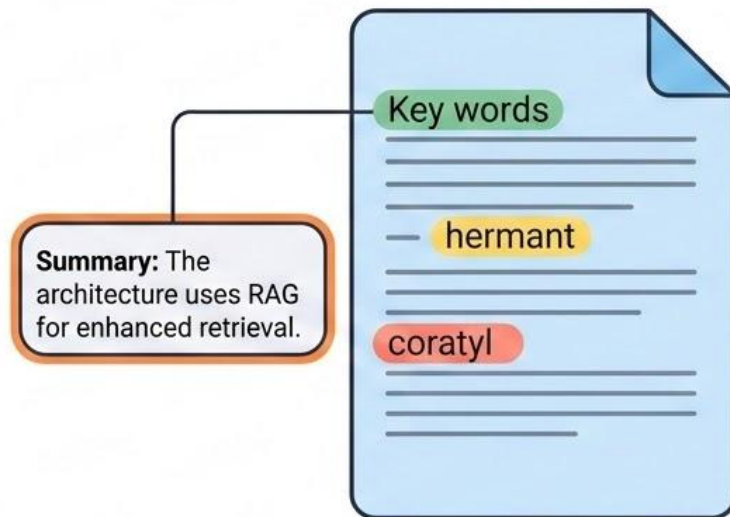
Extraction

- LLM-based extraction



Metadata

- Chunk-level summaries as metadata

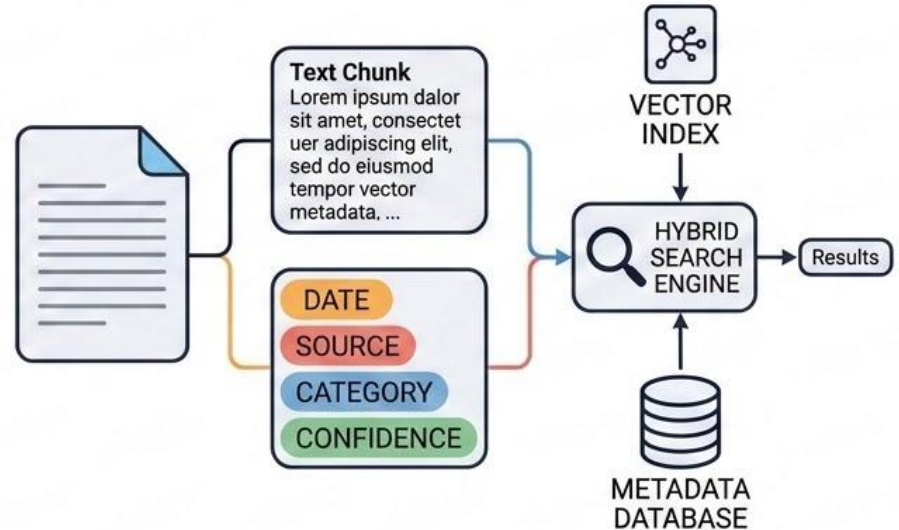


METADATA GENERATION FOR HYBRID SEARCH



IMPORTANCE

- Enables filtering (date, source, category, confidence)
- Powers hybrid search (vector + metadata)
- Improves retrieval precision dramatically



Vector Database Landscape



Managed

- Pinecone, Weaviate Cloud



Self-hosted

- Weaviate, Qdrant, Milvus



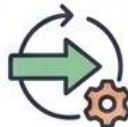
In-process

- Chroma, FAISS



In-Stack

- pgvector, OpenSearch, S3 Vector



Vector DB Decision Matrix

		Type	Ease	Cost	Scale	Use Case
	Pinecone	Managed	✓ Setup	Op Cost	✓ High	Production RAG
	Chroma	Local	✓ High	Zero Cost	● Local	Prototyping
	pgvector PostgreSQL	In-Stack	✓ High	Low Cost	✓ High	PG Ecosystems
	Weaviate / Qdrant	Self-hosted	Moderate	Mod/High	✓ High	Infra Control
	OpenSearch	Scalable	Moderate	Mod/High	Very High	Large-scale workloads

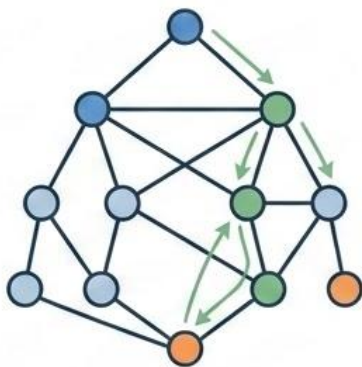
HNSW vs IVF Indexes

HNSW

 Fast search

 High memory

 Great accuracy

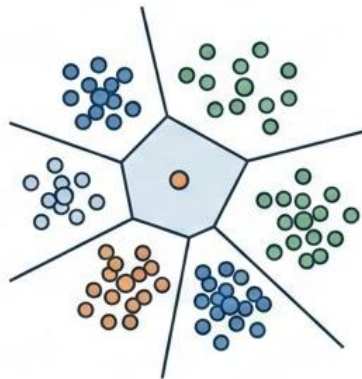


IVF

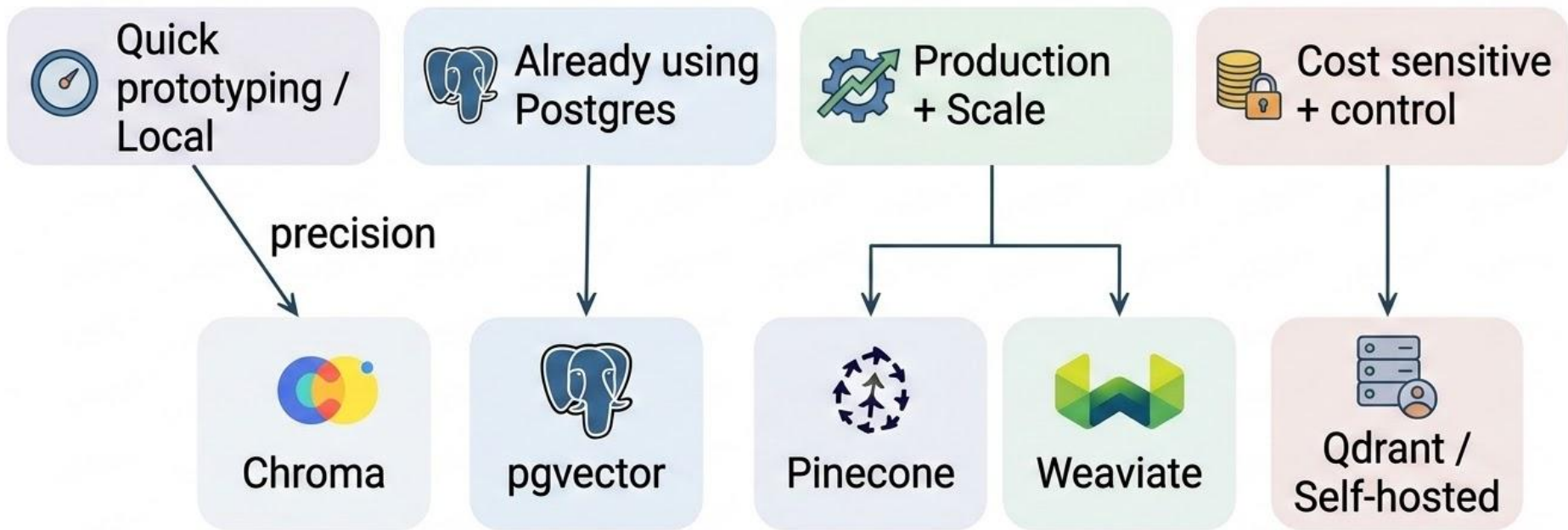
 Lower memory

 Slightly slower

 Good for very large scale



Choosing the Right Vector Database



Best Practices for Chunk Enrichment

- ✓ Always enrich with metadata
- ✓ Redact PII before indexing
- ✓ Keep metadata lightweight
- ✓ Test retrieval quality after enrichment
- ✓ Balance enrichment cost vs performance gain

Key Takeaways

- Raw chunks are not enough — Enrichment is essential
- Metadata turns vector search into hybrid search
- Choose Vector DB based on scale, cost, and control
- Start simple with Chroma, scale thoughtfully

Smart Chunks + Smart Storage = Powerful RAG